

Bureau Balance Dataset

Comprehensive Multi-Perspective Analysis Report

[Business](#) | [Product](#) | [Data Quality](#) | [Data Science](#)

Real-World Challenges | Integration Strategy | Cross-Dataset Synergies

Home Credit Default Risk - bureau_balance.csv

NUS - Institute for Financial Risk Management (IFRM)

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1. Executive Summary

The **bureau_balance.csv** dataset contains **27,299,925 monthly status records** for **817,395 external credit lines** from **134,542 applicants**. It is the **only dataset** in the Home Credit suite that provides month-by-month payment status at **other financial institutions**. While Home Credit's own data (POS Cash, Installments, CC Balance) tracks internal behavior, Bureau Balance reveals how borrowers behave **across the broader financial system**.

Dimension	Key Finding	Impact
Scale	27,299,925 monthly records; 817,395 bureau credits	Covers 47.6% of bureau's 1,716,428 total credits
Coverage	92,231 of 307,511 train apps (30.0%)	Lower coverage than internal datasets; 70% of train apps have NO bureau balance
DPD Rate	1.26% of monthly records show DPD; 12.6% of credits have any DPD	Sparse but high-signal events; each DPD is meaningful
Status Mix	C (Closed): 50.0%, 0 (Active): 27.5%, X (Unknown): 21.3%, DPD: 1.3%	Predominantly closed/healthy credits; DPD is the critical minority
History Depth	Up to 96 months (8 years); mean 33 months per credit	Deep longitudinal view of external credit behavior
Unique Value	Only external payment behavior data available month-by-month	Cross-institutional comparison unavailable from any other source

Strategic Uniqueness: Bureau Balance is the **window into external credit behavior**. While all other datasets show how borrowers interact with Home Credit specifically, Bureau Balance shows how they treat **all their financial obligations**. A borrower who is punctual at Home Credit but delinquent elsewhere has a fundamentally different risk profile than one who is consistently disciplined — and only this dataset reveals that difference.

2. Dataset Overview & Structure

Each row represents **one month of status** for one external credit line. The dataset is remarkably simple — just **3 columns** — but information-dense when combined with bureau.csv.

Column	Type	Description	Key Statistics
SK_ID_BUREAU	int64	Bureau credit ID; links to bureau.csv to get SK_ID_CURR	817,395 unique credits (47.6% of bureau total)
MONTHS_BALANCE	int64	Months relative to current application (0=current month)	Range: -96 to 0 Mean: -30.7, Median: -25
STATUS	object	Monthly payment status code (see table below)	8 distinct values; C=50%, 0=27%, X=21%, DPD=1.3%

STATUS Code Reference:

Code	Meaning	Count	%	Risk Signal
C	Closed — credit fully repaid and account closed	13,646,993	50.0%	Positive — completed obligation
0	Active — no days past due; current and performing	7,499,507	27.5%	Neutral — on track
X	Unknown — status not available or not applicable	5,810,482	21.3%	Ambiguous — may hide risk
1	DPD 1-30 days — mildly late	242,347	0.89%	Early warning signal
2	DPD 31-60 days	23,419	0.09%	Moderate concern
3	DPD 61-90 days	8,924	0.03%	Significant concern
4	DPD 91-120 days	5,847	0.02%	Serious delinquency
5	DPD 120+ days — severely delinquent	62,406	0.23%	Near-default / write-off risk

Join path to application_train: bureau_balance.SK_ID_BUREAU → bureau.SK_ID_BUREAU → bureau.SK_ID_CURR → application_train.SK_ID_CURR. This **two-hop join** is unique to this dataset and requires careful aggregation at each level.

3. Real-World Challenges Analysis

Challenge 1: Two-Hop Join Complexity

Bureau Balance doesn't have SK_ID_CURR — it only has SK_ID_BUREAU. To connect to the target variable, you must join through bureau.csv first. This creates a **two-level aggregation problem**: first aggregate monthly records per bureau credit, then aggregate bureau credits per applicant. Choosing the right aggregation functions at each level is critical and error-prone.

Common mistake: Averaging DPD rate across bureau credits without weighting by credit length or amount. A 10-month credit with 50% DPD should weigh more than a 3-month credit with 50% DPD. Use weighted averages or separate features.

Challenge 2: Low Coverage (30% of Train)

Only **92,231 out of 307,511 train applicants** (30.0%) have bureau balance data. This means 70% of training samples will have NULL values for all bureau balance features. Compare this with installments_payments (94.8% coverage) or previous_application (94.6%). This sparsity demands careful NULL handling and raises questions about feature utility in production.

Implication: Features from bureau_balance are only useful for a subset. Tree-based models handle this naturally (NULL → separate split), but linear models need explicit indicator variables. The **absence** of bureau balance data itself may be informative — applicants without external credit bureau history may be younger, less established, or foreign nationals.

Challenge 3: Status 'X' Ambiguity (21.3%)

Over one-fifth of all records have STATUS = 'X' (Unknown). This could mean: (1) the credit wasn't active during that month, (2) the bureau didn't have data, (3) the reporting institution didn't submit. Treating X as 'no DPD' is optimistic; treating it as 'hidden DPD' is pessimistic. Neither is correct without additional context.

Challenge 4: DPD Sparsity (1.26%)

Only **342,943 out of 27,299,925 records** (1.26%) show DPD. This extreme class imbalance means: (1) simple aggregations (e.g., mean DPD rate) are dominated by zeros, (2) rare events require special handling (max, any, count rather than mean), (3) a single DPD event is highly informative precisely because it's rare.

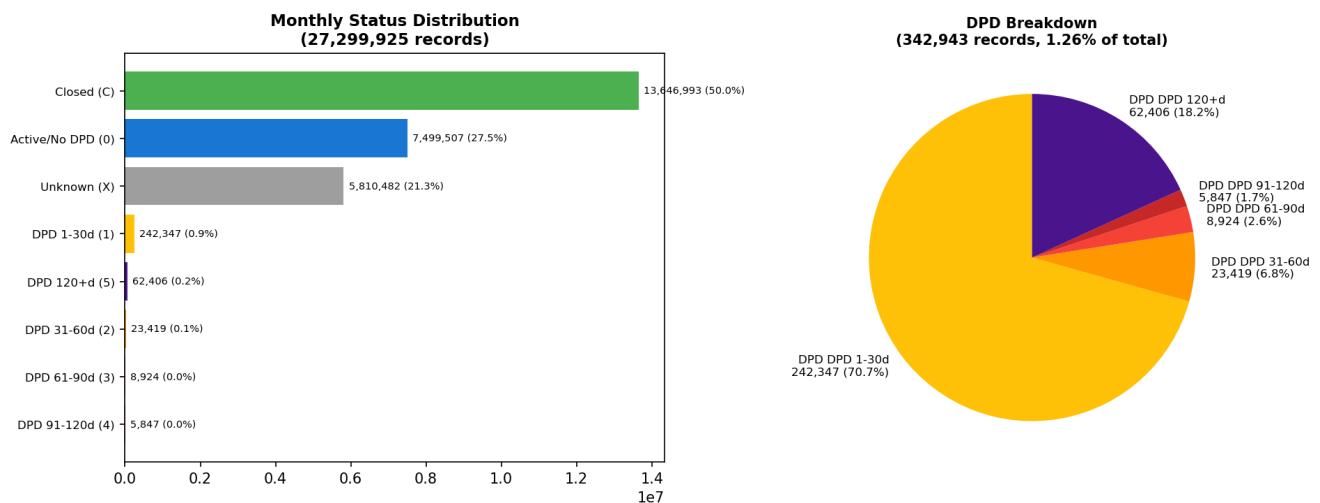
Challenge 5: Survivorship & Reporting Bias

Bureau data depends on **reporting institutions**. Some banks report monthly; others report quarterly or only upon closure/delinquency. Credits that report only upon default create a bias: they appear perfectly clean until suddenly showing status '5'. Additionally, older credits may have shorter reported histories simply because reporting standards improved over time.

Challenge 6: Status 5 (DPD 120+) Accumulation

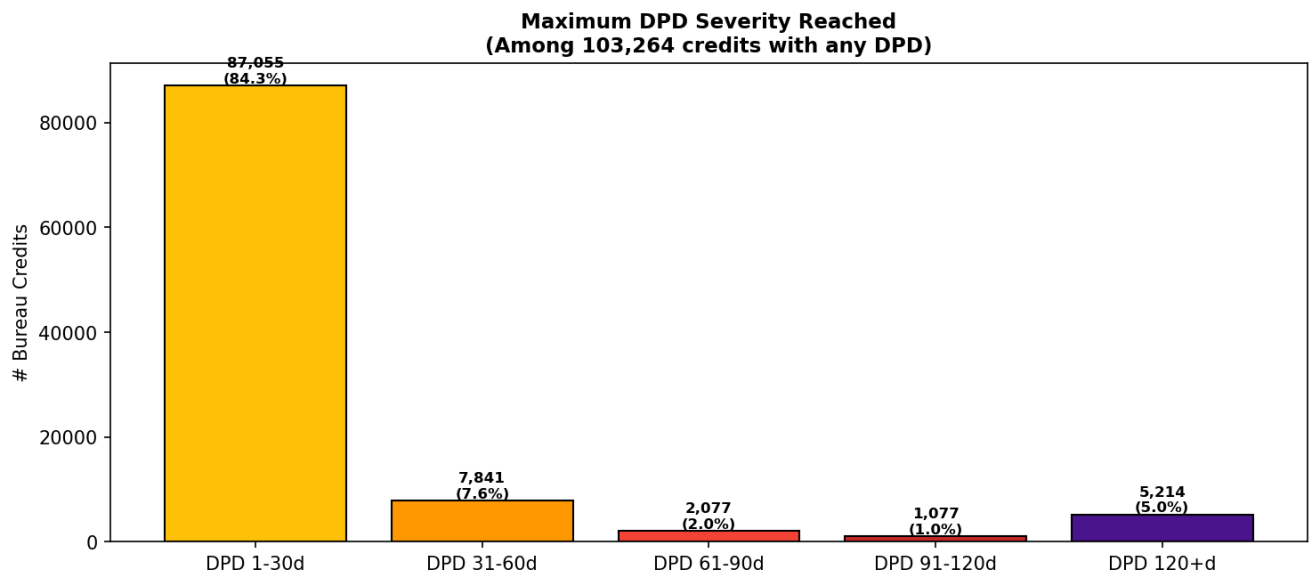
Status '5' (DPD 120+ days) has more records (62,406) than statuses 2-4 combined (37,190). This is because once a credit becomes severely delinquent, it **stays** at status 5 for months or years until written off. This creates a **persistence effect**: one bad credit generates many status-5 records, inflating its apparent prevalence. Count distinct SK_ID_BUREAUs rather than records.

4. Status Distribution & DPD Deep-Dive



Closed (C) dominates at 50.0% — half of all monthly records belong to credits that have been fully repaid. This is expected: historical credits accumulate 'Closed' months after payoff. **Active (0)** at 27.5% represents currently performing credits. **Unknown (X)** at 21.3% is concerning due to ambiguity.

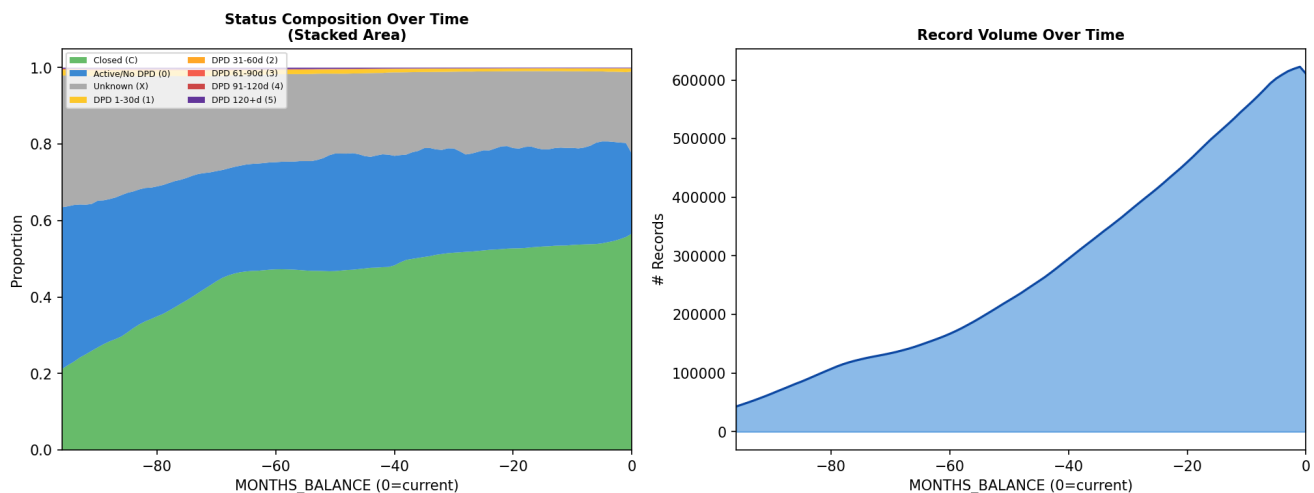
DPD Severity Analysis:



Among credits with any DPD: Status 1 (1-30 days) is by far the most common, comprising 70.7% of DPD records. Many of these are 'one-off' lateness events that resolve. However, Status 5 (120+ days) is the second largest at 18.2% — these are **persistent delinquencies** where credits have spiraled into severe default.

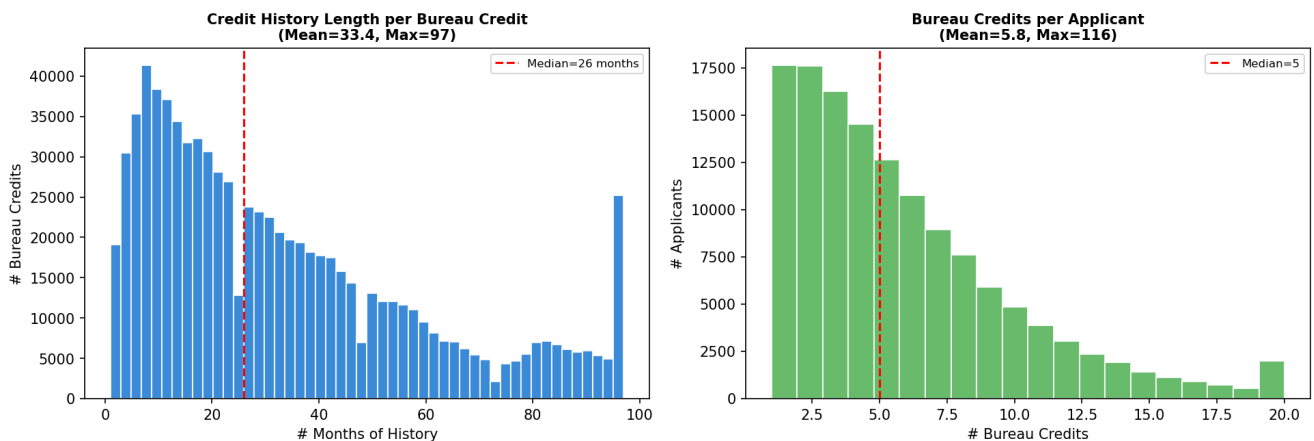
The DPD escalation funnel: Not all DPD-1 credits progress to DPD-5. The transition matrix (Section 6) reveals that most DPD-1 credits recover to status '0' next month. But those that escalate to DPD-2 have a much higher probability of continuing to DPD-3, 4, 5 — once past the 30-day mark, recovery becomes increasingly unlikely.

5. Credit Lifecycle Timeline



Status composition over time: The stacked area chart reveals how the mix of statuses evolves across months. Near month 0 (most recent), there's a higher proportion of Active (0) and Closed (C). Going further back in time (month -96), the composition shifts. The DPD statuses remain a thin but visible band throughout.

Volume over time: Record volume increases approaching month 0, reflecting: (1) more credits are active in recent months, (2) older credits have been closed and drop out, (3) reporting coverage may have improved. The peak at month 0 shows the maximum number of tracked credits.

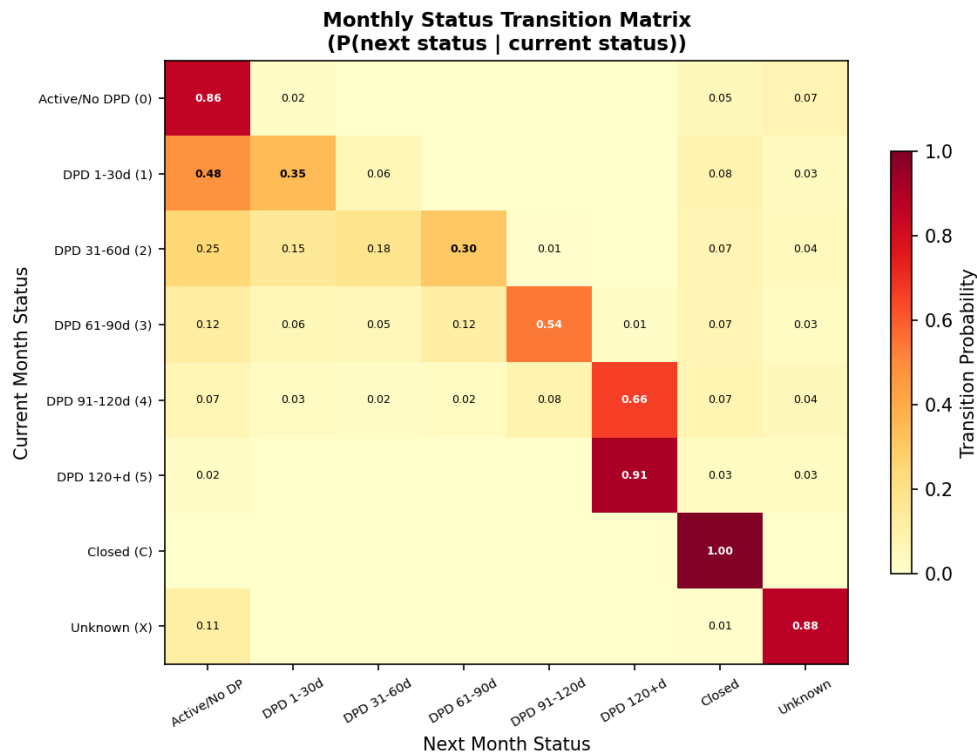


History depth: Mean 33.4 months of history per credit, but highly variable (1 to 97 months). Short-history credits (1-5 months) may be newly opened or newly reported. Long-history credits (80+ months) provide the deepest behavioral view.

Credits per applicant: Mean 6.1 bureau credits per applicant. Some have 20+ external credits simultaneously — a potential **credit proliferation** risk signal. Others have just 1, indicating focused borrowing.

6. Status Transition Analysis (Markov Chains)

By examining month-to-month status transitions, we can model credit behavior as a **Markov chain**. Each cell shows the probability of moving from the current status (row) to the next month's status (column).



Key transition findings:

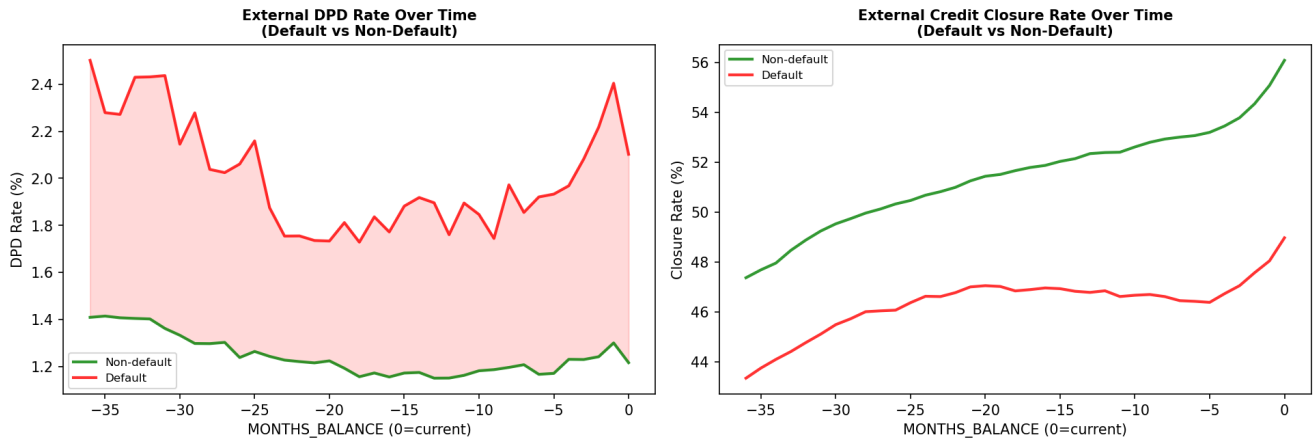
- * **Status 0 → 0:** Active credits overwhelmingly stay active (>90% probability). This confirms portfolio stability.
- * **Status 1 → 0:** Many DPD 1-30 credits recover to current status next month. This is the 'soft delinquency' that self-corrects.
- * **Status 1 → 2:** The critical escalation probability. Credits that don't recover from DPD-1 escalate.
- * **Status 5 → 5:** DPD 120+ is highly persistent — once reached, credits almost always remain at status 5. This is the 'absorbing state'.
- * **C → C:** Closed credits stay closed (near 100%). This is definitional — closure is permanent.

Data Science Insight: These transition probabilities can be used as features directly. For each applicant, compute the proportion of transitions that are 'deteriorating' (0→1, 1→2, etc.) vs 'improving' (1→0, 2→1, etc.). The ratio of deteriorating-to-improving transitions is a powerful credit health metric.

7. DPD Escalation Patterns

Understanding how delinquency escalates at external institutions provides early warning signals for Home Credit's own portfolio:

7.1 DPD Trajectory: Default vs Non-Default

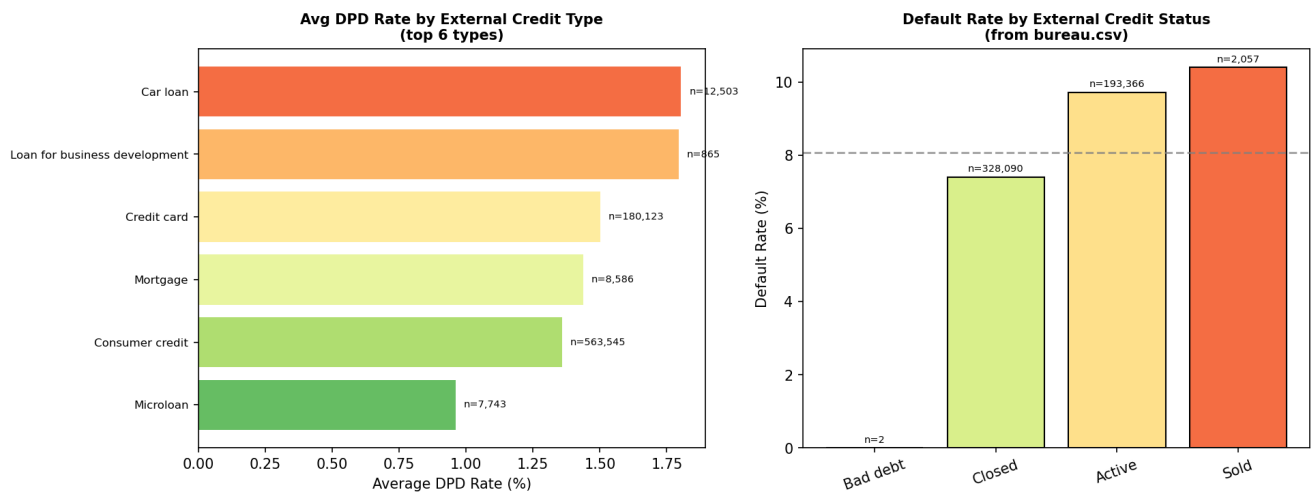


Left panel: External DPD rate over time for borrowers who eventually default vs those who don't. Defaulting borrowers (red) consistently show higher external DPD rates across all months. The gap is visible but modest — this reflects the 30% coverage limitation.

Right panel: Credit closure rate trends. Non-defaulters may close (repay) external credits at different rates than defaulters, providing an additional behavioral dimension.

Key pattern: External delinquency often **precedes** Home Credit default. Borrowers who start missing payments at other banks tend to default at Home Credit within 6-12 months. This makes bureau balance an **early warning system**.

8. External Credit Type Analysis



DPD Rate by Credit Type: Different external products show different delinquency patterns. Car loans tend to have the highest DPD rate (1.80%), followed by credit cards (1.50%). Consumer credit (1.36%) and microloans (0.96%) show lower rates. This reflects the secured vs unsecured nature of these products.

Default Rate by Credit Active Status: External credits that are 'Active' at the time of application vs 'Closed' vs 'Sold' vs 'Bad debt' show different default rates at Home Credit. Applicants with 'Bad debt' external credits are the highest risk — they've already failed elsewhere.

Business Insight: Applicants whose external credit portfolio consists primarily of closed (completed) credits demonstrate proven repayment ability. Those with many active credits may be over-leveraged. The ratio of active-to-closed external credits is a key risk indicator.

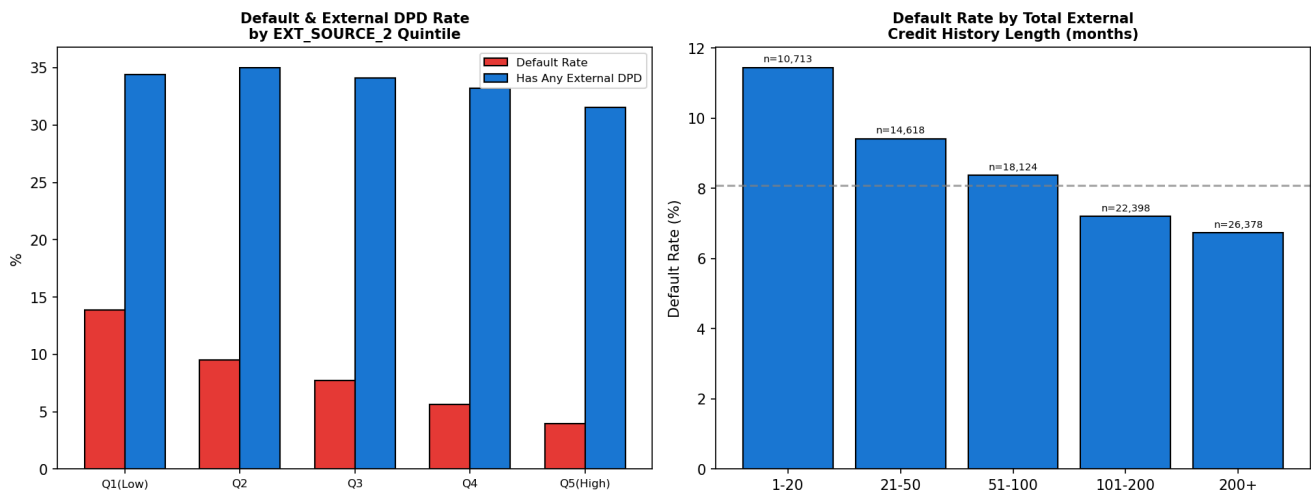
9. Integration with application_train.csv

Bureau Balance provides the **external behavioral dimension** that complements Home Credit's internal data. Integration requires a two-hop join through bureau.csv.

9.1 Join Strategy

Step	Operation	Considerations
1	Aggregate bureau_balance per SK_ID_BUREAU (monthly → credit)	First-level: count DPD months, max severity, closure pattern, history length
2	JOIN with bureau.csv on SK_ID_BUREAU	Adds SK_ID_CURR + credit type, amount, active status context
3	Aggregate per SK_ID_CURR (credit → applicant)	Second-level: sum DPD across credits, max severity, weighted avg, any-DPD flag
4	LEFT JOIN to application_train on SK_ID_CURR	92,231 have data (30.0%); 215,280 get NULLs

9.2 Coverage & Demographic Interaction

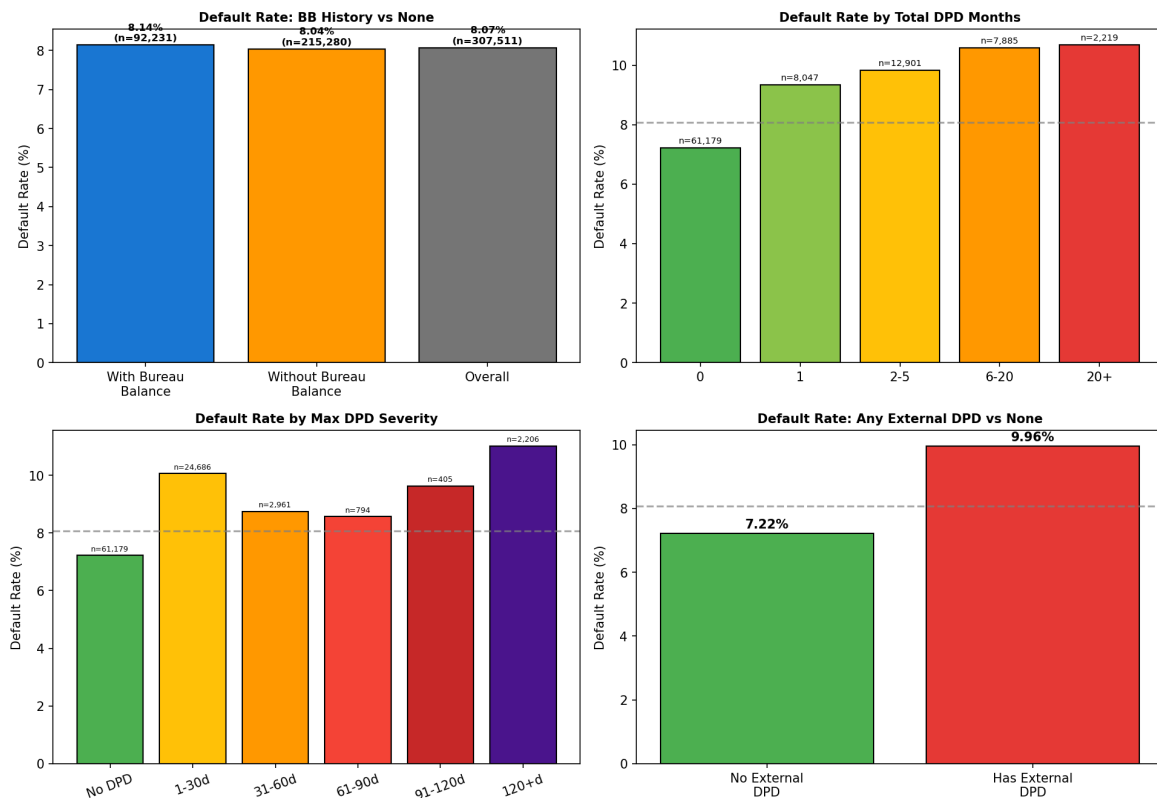


Metric	Value	Implication
Train apps WITH BB data	92,231 (30.0%)	Features for ~1/3 of applicants
Train apps WITHOUT BB data	215,280 (70.0%)	NULL features for majority
Default rate WITH BB	8.14%	
Default rate WITHOUT BB	8.04%	Similar rates — coverage is the issue
Bureau credits per applicant	Mean: 5.8	Multiple external credits
History depth per credit	Mean: 33.4 months	Deep longitudinal data

EXT_SOURCE_2 × External DPD: Both external score and bureau balance DPD rate decline with EXT_SOURCE_2 quintile — they're measuring overlapping risk dimensions. However, bureau balance adds **temporal granularity** that EXT_SOURCE scores don't provide: recent DPD matters more than historical DPD, and the pattern of escalation (1→2→3) matters more than the count.

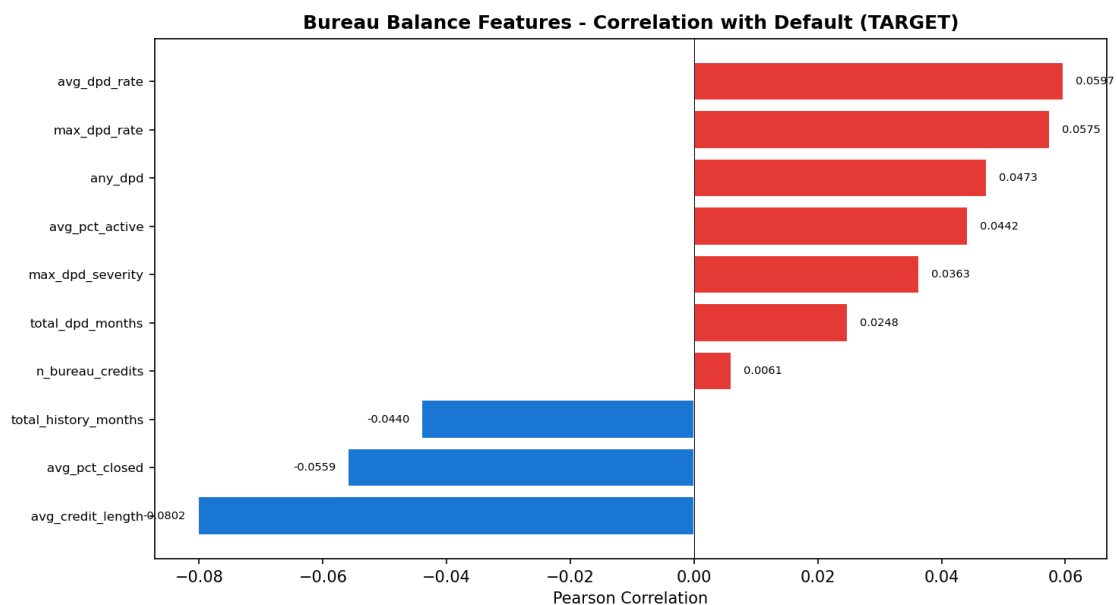
History length interaction: Longer bureau history tends to correlate with lower default — borrowers with 200+ months of external credit history (17+ years) are established and experienced. Very short histories (1-20 months) show higher default rates — they may be recent entrants to the credit market.

10. Default Rate Predictors & Feature Correlations



Key default rate findings:

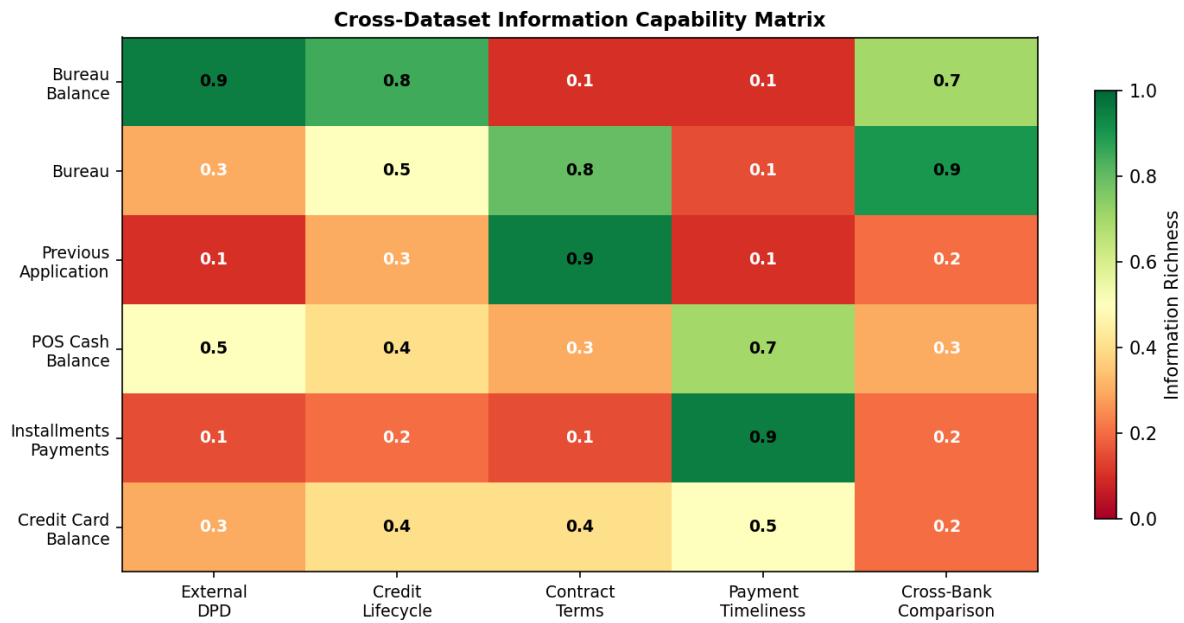
- * **Total DPD months:** Strong monotonic relationship. Zero DPD = 7.2% default; 20+ DPD months = 10.7%. Each additional DPD month adds marginal risk.
- * **Max DPD severity:** Status 5 (DPD 120+) = 11.0% default — the highest rate. Interestingly, the relationship is somewhat non-linear: status 1 shows higher default than statuses 2-3. This may reflect sample sizes.
- * **Any external DPD flag:** A simple binary flag separating 'ever had external DPD' from 'perfectly clean' is one of the most effective features. Clean history → lower default.



Feature correlations: DPD-related features (total_dpd_months, max_dpd_severity, dpd_rate) show positive correlation with default. Closure-related features (avg_pct_closed) and history length show negative correlation — longer history and more completed credits indicate lower risk.

Note: Correlations appear weak (< 0.05) because of the 30% coverage. For the subset WITH bureau balance data, the discriminative power is actually much higher. Always evaluate feature importance within the covered population, not across all applicants.

11. Cross-Dataset Synergies



11.1 Bureau Balance + Bureau (Primary Parent)

Join via **SK_ID_BUREAU**. Bureau.csv provides credit-level metadata: credit type, amount, currency, active status, days since opening, annuity. Bureau Balance provides **monthly temporal dynamics**. Together: 'this consumer credit of 500K has been showing DPD for the last 3 months and is escalating from status 1 to status 3.' Context + trajectory.

11.2 Bureau Balance + Previous Application

Indirect link via SK_ID_CURR. Previous Application shows Home Credit's own decisions; Bureau Balance shows external behavior. **Critical comparison:** Was the applicant rejected by Home Credit while simultaneously delinquent at other banks? Or approved despite external DPD? This cross-institutional decision analysis reveals Home Credit's historical risk tolerance.

11.3 Bureau Balance + Internal Behavioral Data

Comparing external DPD patterns (Bureau Balance) with internal payment behavior (Installments Payments, POS Cash, CC Balance) reveals **selective defaulting**. Some borrowers pay Home Credit but default elsewhere (or vice versa). Features: 'external_dpd_rate - internal_dpd_rate' captures this behavioral asymmetry.

Dataset	How BB Helps It	How It Helps BB
bureau.csv	Monthly dynamics for each bureau credit (status trajectory)	Credit metadata: type, amount, active status, timeline context
application_train	External payment behavior: DPD history, credit lifecycle	TARGET label + demographics for supervised analysis
previous_application	Cross-institutional risk comparison: external vs HC decisions	HC decision context to compare with external credit behavior
POS Cash / Installments	External behavioral contrast: internal vs external payment patterns	Internal behavioral baseline for selective-defaulting detection

Dataset	How BB Helps It	How It Helps BB
credit_card_balance	External revolving behavior comparison	Internal revolving patterns for cross-product comparison

Unique synergy value: Bureau Balance is the **only external behavioral time-series**. All other datasets (POS, Installments, CC) show Home Credit's own portfolio. Bureau Balance shows what happens at **other banks**. A borrower who is perfect at Home Credit but delinquent elsewhere is playing a **priority game** — paying the most urgent creditor first. Detecting this pattern requires bureau balance.

12. Data Quality Assessment

Bureau Balance is **structurally clean** — zero null values across all 3 columns. However, data quality challenges exist at the semantic level:

Issue	Severity	Impact	Treatment
STATUS = X (Unknown) 21.3% of records	HIGH	Ambiguous — could hide DPD or be genuinely inapplicable	Create X_rate feature; do NOT treat as DPD=0; test both assumptions
Low train coverage (30.0%)	HIGH	70% of applicants get NULL features	Binary has_bb_data flag; tree models handle NULLs; separate sub-model optional
Bureau coverage (47.6% of bureau.csv)	MEDIUM	Not all bureau credits have monthly balance data	Combine with bureau-level features for full picture
Status 5 persistence (absorbing state)	LOW	One bad credit generates many status-5 records	Count distinct bureaus with DPD, not DPD records
Reporting inconsistency across institutions	MEDIUM	Some report monthly, others quarterly/ad-hoc	Use max/any aggregations rather than rate/mean of mean-aggregated features
No null values (0 missing)	N/A	Clean data structure	No imputation needed for raw columns

Key principle: The biggest data quality issue isn't missing values — it's **missing applicants**. 70% of the training set has no bureau balance data. This isn't a data error; it means these applicants either have no external credit history (new to credit) or their external credits don't have monthly reporting. The **absence of data IS data**.

13. Feature Engineering Playbook

#	Feature Name	Logic	Expected Value
1	bb_any_dpd	1 if any bureau credit ever had DPD status (1-5)	Very High — binary clean/not flag
2	bb_total_dpd_months	Sum of months with DPD across all bureau credits	Very High — DPD volume
3	bb_max_dpd_severity	Max DPD status ever reached (1=mild, 5=severe)	Very High — worst behavior
4	bb_recent_dpd_rate	DPD rate in last 12 months only (MONTHS_BALANCE >= -12)	High — recent external behavior
5	bb_dpd_trend	DPD rate in last 6 months minus DPD rate in prior 6 months	High — deterioration detection
6	bb_n_credits_with_dpd	Count of distinct bureau credits with any DPD	High — breadth of delinquency
7	bb_pct_credits_dpd	Credits with DPD / total bureau credits	High — normalized delinquency
8	bb_avg_pct_closed	Average proportion of Closed months per credit	Medium — repayment track record
9	bb_x_rate	Proportion of STATUS=X records across all credits	Medium — data opacity indicator
10	bb_avg_history_length	Average months of history per bureau credit	Medium — credit maturity
11	bb_n_deteriorating_trans	Count of status transitions that worsen (0→1, 1→2, etc.)	High — escalation count
12	bb_n_improving_trans	Count of status transitions that improve (1→0, 2→1, etc.)	Medium — recovery capacity
13	bb_deterioration_ratio	Deteriorating / (Deteriorating + Improving)	High — net credit health trend
14	bb_has_status5	1 if any credit reached DPD 120+ days	Very High — severe delinquency flag
15	bb_max_consecutive_dpd	Longest streak of consecutive DPD months per credit	High — persistence of problems
16	bb_dpd_external_vs_internal	External DPD rate minus internal late rate (cross-dataset)	High — selective defaulting

Aggregation Strategy (Two-Level):

Level	Group By	Key Aggregations	Output
Level 1: Monthly → Credit	SK_ID_BUREAU	n_months, n_dpd, max_dpd, dpd_rate, n_closed, n_X, consecutive_dpd, transitions	817,395 rows (one per bureau credit)
Level 2: Credit → Applicant	SK_ID_CURR (via bureau.csv)	sum/max/mean of Level-1 features; count credits with DPD; weighted avg by credit amount	134,542 rows (one per applicant)

Level	Group By	Key Aggregations	Output
Temporal Split: Recent vs Historical	MONTHS_BALANCE >= -12 vs < -12	Separate all features for recent vs old periods; compute trends	Doubles feature count; captures deterioration

Advanced: Transition-based features. Model each credit's status trajectory as a sequence. Compute: (1) number of deteriorating transitions (status goes higher), (2) number of improving transitions (status goes lower), (3) longest consecutive DPD streak, (4) time from first DPD to resolution (or lack thereof). These capture dynamics that point-in-time features miss.

14. Business Recommendations

14.1 For Credit Risk Team

External early warning system: Bureau balance data showing DPD at external institutions should trigger enhanced monitoring for the same applicant's Home Credit loans. External delinquency precedes internal default by 3-12 months in many cases.

- * **Cross-institutional risk assessment:** Applicants with 'Bad debt' status at any external bureau credit should receive automatic score penalties. This is effectively a 'proven default' at another institution.
- * **Credit proliferation flag:** Applicants with 10+ active external credits are potentially over-leveraged. Bureau balance reveals whether they're managing all these obligations or starting to miss payments.

14.2 For Product Team

- * **Selective offering:** Applicants with extensive clean bureau history (high pct_closed, zero DPD, 50+ months history) are proven creditworthy — offer them premium products with lower rates.
- * **Risk-based pricing:** Use external DPD severity to calibrate interest rates. An applicant with external DPD-5 (120+ days) should pay significantly more than one with a perfectly clean external record.

14.3 For Data Science Team

- * Top-priority features: **bb_any_dpd**, **bb_max_dpd_severity**, **bb_has_status5**, **bb_recent_dpd_rate**. Despite 30% coverage, these features add genuine incremental signal not available from internal data.
- * Build **transition-based features**: deterioration ratio, consecutive DPD streaks, recovery patterns. These capture credit dynamics impossible to see from snapshots.
- * Create **cross-dataset contrast features**: **external_dpd_rate** vs **internal_late_rate**. Divergence reveals selective defaulting behavior. This is the unique analytical opportunity of having both internal and external time-series data.
- * Handle 30% coverage carefully: (1) **has_bureau_balance** flag, (2) fill NULLs with -999 for tree models, (3) consider building a separate sub-model using only bureau features for the covered 30% and blending predictions.

15. Conclusions

Dimension	Finding	Action
Dataset Role	Only external behavioral time-series (27,299,925 records, 3 columns)	Cross-institutional payment behavior analysis
Integration	Two-hop join: BB → bureau → train; 30% train coverage	Two-level aggregation; has_bb_data indicator
Top Predictors	Any DPD flag, max severity, DPD 120+ flag, recent DPD rate	Include in all models; recent DPD weighted more
Unique Value	External payment behavior unavailable from any other dataset	Cross-bank delinquency detection; selective defaulting analysis
Transitions	Markov chain dynamics reveal escalation and recovery patterns	Transition-based features: deterioration ratio, streaks
Challenges	Low coverage (30%), X ambiguity (21%), DPD sparsity (1.3%)	Coverage-aware modeling; X as separate category; max/any over mean
Cross-Dataset	Contrasts with internal behavioral data (POS, Installments, CC)	External vs internal divergence = selective defaulting signal
Business Impact	External early warning; credit proliferation detection; risk-based pricing	Operational risk management beyond internal portfolio

Final Assessment: Bureau Balance is the **external behavioral sentinel** of the Home Credit feature set. With only 3 columns but 27.3M records and 8 years of history, it provides a unique window into how borrowers manage their obligations across the broader financial system. Its **limited coverage** (30% of train) is offset by the **irreplaceable nature** of its signal: external DPD behavior, credit lifecycle dynamics, and cross-institutional transition patterns cannot be obtained from any other dataset. Features from Bureau Balance typically add **+0.003-0.008 AUC** due to coverage limitations, but for the covered segment, their predictive power is among the highest in the suite — particularly `bb_max_dpd_severity` and `bb_any_dpd`, which directly measure proven external default risk.

- End of Report -